

Color Transfer Notes

Reference Image Selection

Reference image needs: - The **color tones** you want (warm, cool, vibrant, muted)
- Good **color variety** (not flat/monotone) - Quality (sharpness, resolution)
doesn't matter - only color statistics are used

How Reference Affects Output

Reference Property	Effect on Output
Vibrant colors (high std)	Output more vibrant
Muted colors (low std)	Output more muted
Warm tones (high A mean)	Output warmer
Cool tones (low A mean)	Output cooler

Reinhard Method (LAB Space)

Formula: $(\text{pixel} - \text{target_mean}) / \text{target_std} * \text{ref_std} + \text{ref_mean}$

- Shifts input colors to match reference color distribution
- Uses global statistics (mean & std) per channel
- Same formula applied to all pixels

LAB Channels

Channel	Represents	Range
L	Lightness	0 (black) to 100 (white)
A	Green to Red	-128 (green) to +128 (red)
B	Blue to Yellow	-128 (blue) to +128 (yellow)

NILUT vs Reinhard

Aspect	Reinhard	NILUT
Approach	Global stats	Neural network
Formula	Same for all pixels	Learned per-pixel
Speed	Fast	Slower
Consistency	High	Variable